Distinguishing Exponential from Linear Change
Answer Key

Algebra 1   Exponential Growth and Decay

Quick Answers

1. 23	4. 768	7. 5	10. 4242.88
2. 972	5. 409.38	8. 665.57	
3. 400	6. 4, 2.25	9. 2, 16000	

Curriculum

Level: Algebra 1   Exponential Growth and Decay
HSF-BF.A.2, HSF-LE.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

5. *If they got 400.0: multiplied by 1.5 once and by 5 instead of raising 1.5 to the 5th power*
8. *If they got -2400.0: treated 12 percent per year as 12 percent of the original price every year*
10. *If they got -999.62: multiplied the starting cent by 2 and by the number of days instead of doubling repeatedly*

1. Table 5, 8, 11, 14 — constant difference.

Step 1. Test both patterns before naming one. Differences: $8-5=3$, $11-8=3$, $14-11=3$ — constant. Ratios: $8/5=1.6$ but $11/8=1.375$ — not constant.
Step 2. A constant difference is the signature of linear change, so $f(n)=5+3n$ with $a=5$ and $d=3$.
Step 3. $f(6)=5+3\cdot 6=5+18$. $\boxed{23}$
Read the table, not the shape: four increasing values could be either model. Only the difference row settles it.

2. Table 4, 12, 36, 108 — constant ratio.

Step 1. The differences 8, 24, 72 are growing, which rules out linear change, so test the ratios: $12/4=3$, $36/12=3$, $108/36=3$ — constant.
Step 2. A constant ratio is the signature of exponential change: $g(n)=4\cdot 3^n$ with $a=4$ and $r=3$.
Step 3. $g(5)=4\cdot 3^5=4\cdot 243$. $\boxed{972}$
Note: the differences of an exponential pattern grow in the same ratio — 8, 24, 72 also triples. Growing differences are the fingerprint of a constant ratio.

3. Rosa's savings: 200 dollars plus 25 dollars a month.

Step 1. Classify from the verb before computing. "Adds 25" is repeated addition of a fixed amount, so the balance changes by a constant difference — linear, not exponential. No interest means nothing is multiplied.

Step 2. The rule is $B(n) = 200 + 25n$, so $B(8) = 200 + 25 \cdot 8 = 200 + 200$. $\boxed{400}$

Contrast: an account paying 25 percent interest would multiply by 1.25 each month and reach a far larger balance — same-looking numbers, different operation.

4. 6 cells doubling every hour.

Step 1. "Doubles" means multiply by 2, and a repeated multiplication by a fixed factor is exponential change with $r = 2$.

Step 2. Starting at $a = 6$, the rule is $C(n) = 6 \cdot 2^n$.

Step 3. $C(7) = 6 \cdot 2^7 = 6 \cdot 128$. $\boxed{768}$

Why not linear: the colony adds 6 cells in the first hour but 384 in the seventh. The amount added is not fixed — the multiplier is.

5. Plan L versus Plan E, both starting at 100 dollars.

Step 1. Plan L adds, so $L(n) = 100 + 50n$; Plan E multiplies, so $E(n) = 100 \cdot 1.5^n$. Write both rules first — the comparison is between rules, not between two numbers.

Step 2. Part (b): Plan E is behind early because 50 percent of a small balance is small, but the amount E adds grows every step while L's stays at 50. Once E's step passes 50 dollars it gains ground and never gives it back.

Step 3. At step 5: $E(5) = 100 \cdot 1.5^5 = 100 \cdot 7.59375 = 759.375$ dollars, and $L(5) = 100 + 250 = 350$ dollars. The gap is $759.375 - 350$. $\boxed{409.38}$

Watch out: computing $100 \cdot 1.5 \cdot 5 = 750$ instead of $100 \cdot 1.5^5$ gives a gap of 400 dollars. Multiplying by the number of steps is the linear move; exponential change raises the factor to that power.

6. Table 2, 8, 18, 32 — neither model.

Step 1. Part (a): the first ratio is $h(2) \div h(1) = 8 \div 2$. $\boxed{4}$

Step 2. Part (b): the next ratio is $h(3) \div h(2) = 18 \div 8$, which is $\frac{9}{4}$. $\boxed{2.25}$

Step 3. Part (c): 4 and 2.25 are different, so there is no constant ratio and h is not exponential. The differences 6, 10, 14 are not constant either, so h is not linear. It is neither — in fact $h(n) = 2n^2$.

Why this problem matters: "not linear" does not mean "exponential." Both rows have to be tested, and a pattern is allowed to fail both.

7. Linear pattern through $(2, 17)$ and $(5, 32)$.

Step 1. For a linear pattern the rate of change is the same between any two points, so compute it from the pair given rather than assuming consecutive inputs.

Step 2. $\dfrac{32 - 17}{5 - 2} = \dfrac{15}{3}$. $\boxed{5}$

Step 3. Part (b): back up from $(2, 17)$ by two steps of 5 to get $f(0) = 7$, so $f(n) = 7 + 5n$. Check: $f(5) = 7 + 25 = 32$.

Part (c): no. An exponential function through those points has a constant *ratio*, and its rate of change grows from one interval to the next — which is exactly what distinguishes it from this line.

8. A 24000-dollar car: 12 percent a year versus 2400 dollars a year.

Step 1. Dealer A takes a percent of the *current* value, so each year multiplies by $1 - 0.12 = 0.88$ — exponential decay. Dealer B subtracts a fixed amount, so it is linear. Classifying first prevents the usual error.
Step 2. $A(n) = 24000 \cdot 0.88^n$ and $B(n) = 24000 - 2400n$.
Step 3. At $n = 5$: $A(5) = 24000 \cdot 0.88^5 = 24000 \cdot 0.5277319 = 12665.57$ dollars, and $B(5) = 24000 - 12000 = 12000$ dollars. The difference is $12665.566 - 12000$. $\boxed{665.57}$
Watch out: reading "12 percent a year" as 12 percent of the original price every year gives $24000(1 - 0.60) = 9600$ dollars and makes Dealer A look 2400 dollars *worse*. That single misreading turns an exponential model into a linear one and flips the answer's sign.

9. Exponential population: 500 at time 0, 4000 at time 3.

Step 1. Part (a): exponential means $P(n) = 500 \cdot r^n$, so the given pair becomes one equation in r: $500r^3 = 4000$, hence $r^3 = 8$. $\boxed{r = 2}$
Step 2. Part (b): the rule is $P(n) = 500 \cdot 2^n$, so $P(5) = 500 \cdot 32$. $\boxed{16000}$
Compare with linear: a linear model through the same two points would add $\frac{4000-500}{3} \approx 1166.7$ per step and reach about 6333 at time 5 — far less than 16000. Same two points, wildly different futures.

10. Challenge: flat 1000 dollars a day versus a doubling cent.

Step 1. Part (a): day 1 pays 0.01 dollars and has been doubled zero times, so the exponent counts the days *after* day 1: $D(n) = 0.01 \cdot 2^{n-1}$. The flat plan is $F(n) = 1000$, a constant — the extreme case of linear, with $d = 0$.
Step 2. Part (b): on day 10, $D(10) = 0.01 \cdot 2^9 = 5.12$ dollars, so the flat plan pays far more that day. The doubling plan is still tiny.
Step 3. Part (c): on day 20, $D(20) = 0.01 \cdot 2^{19} = 0.01 \cdot 524288 = 5242.88$ dollars, and the flat plan pays 1000 dollars, so the gap is $5242.88 - 1000$. $\boxed{4242.88}$
The whole point: between day 10 and day 20 the doubling plan went from 1000 times behind to 4 times ahead, without changing its rule. A constant ratio always overtakes a constant difference eventually — the only question is when.
Watch out: $0.01 \cdot 2 \cdot 19 = 0.38$ dollars is the linear misreading, and it leaves the doubling plan 999.62 dollars behind.